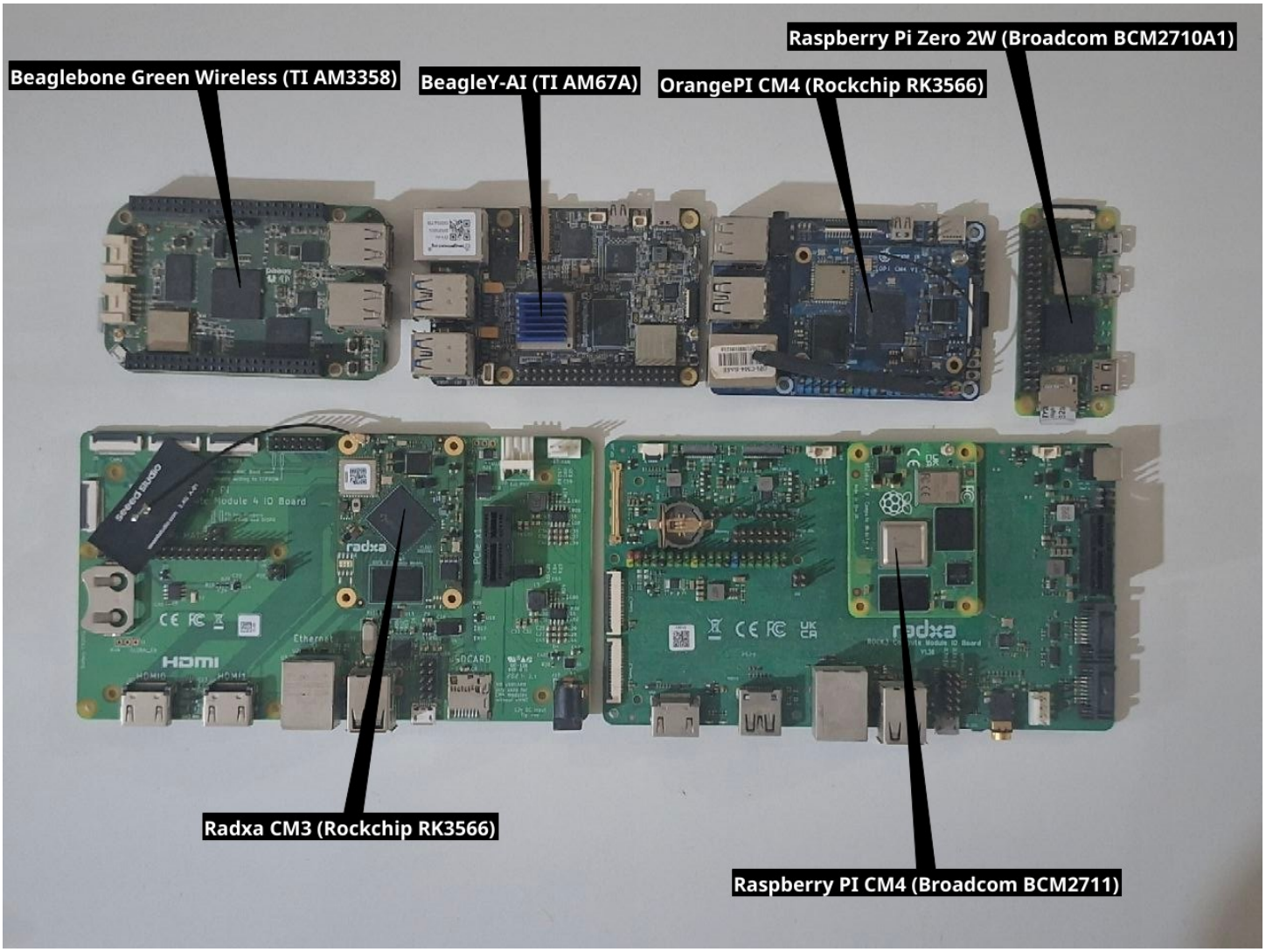
Nazib Abrar

Technical Attachment: Projects, Platforms, and Verification

Embedded Linux Platforms

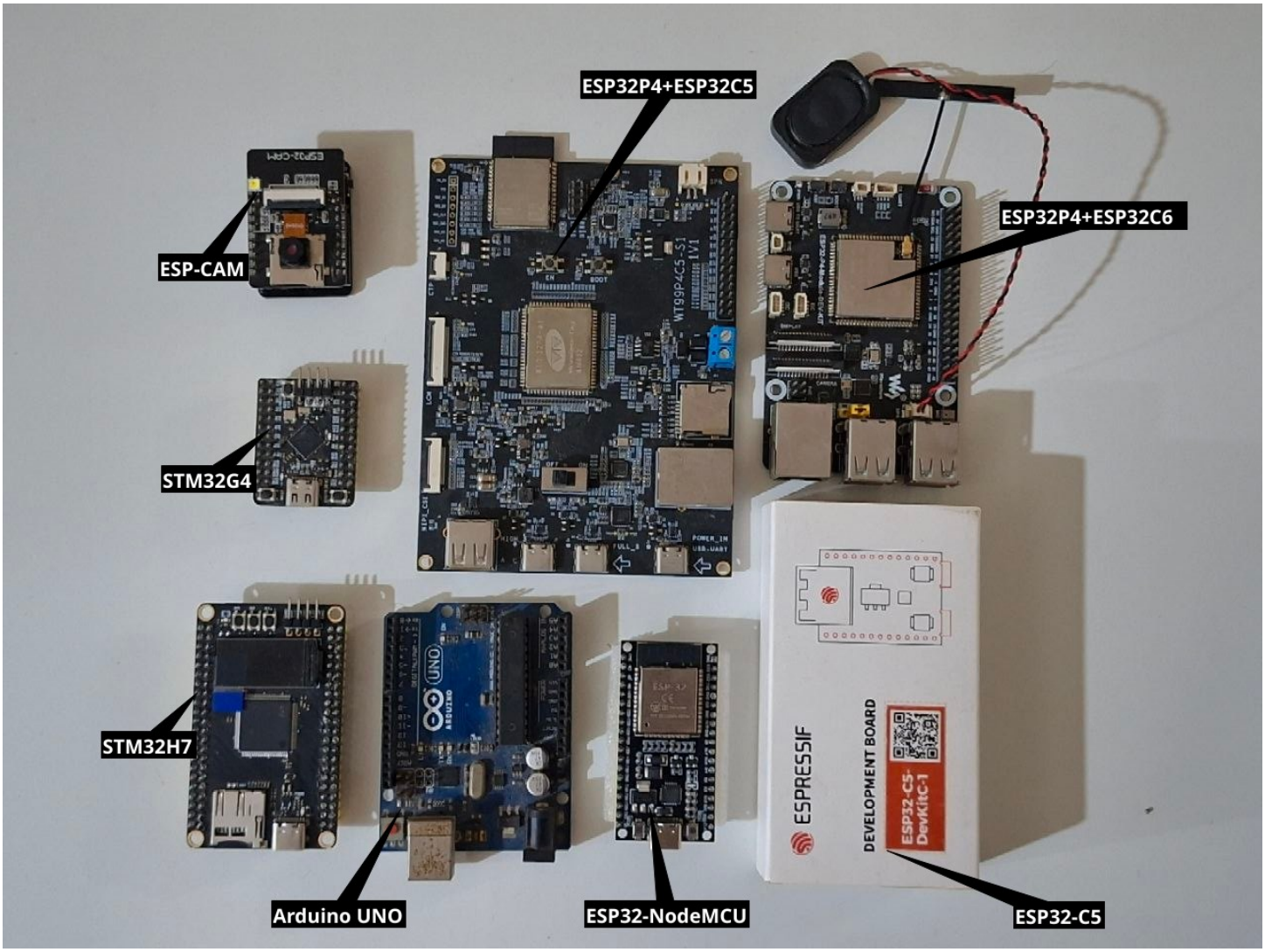
- **OrangePi CM4 and Radxa CM3** (Rockchip RK3566). Client project comprising custom carrier board development, an RKNN based computer vision stack, and a hardware accelerated GStreamer pipeline, deployed on a custom Yocto distribution built for both boards.
- **Raspberry Pi CM4 and Raspberry Pi Zero 2W** (Broadcom BCM2711 and BCM2710A1). Client projects covering a security delivery locker with WebRTC video conferencing and hardware accelerated video recording, including GStreamer pipeline development using `v4l2src`.
- **BeagleY-AI and BeagleBone Green Wireless** (Texas Instruments AM67A and AM3358). Conducted tutoring sessions on the Texas Instruments SDK at the RUET IoT Club, covering device driver development and the AI ISP and NPU hardware acceleration SDKs.



Single board computers and compute modules used in the projects described above.

Microcontroller Platforms

- **STM32H7 and STM32G4**. Client project comprising firmware development for a CAN analyzer capable of inspecting a live CAN bus and streaming its traffic for real time data visualization.
- **ESP32-P4 with ESP32-C5, and ESP32-P4 with ESP32-C6**. Client project implementing non-AI video calling over WebRTC using `esp-peer`.
- **ESP32-CAM**. Employed as the camera node across a range of projects.
- **Interfaces and toolchains**. Practical experience with the SPI, I2C, and CSI camera interfaces, and firmware development in STM32CubeIDE, STM32CubeMX, ESP-IDF, and the Arduino IDE.



Microcontroller development boards used for firmware and peripheral driver work.

Project Demonstrations

SmartSchoolBus (smartschoolbus.gov.bd), a Government of Bangladesh School Bus Safety Platform

- National television coverage of the deployed product by Somoy TV: youtu.be/vm3ZXsQ_LZ8
- Operational walkthrough of the platform, presented in the capacity of lead engineer on the project: youtu.be/xTgJycR2Adk

Vision-Based Autonomous Drone Navigation in GPS-Denied Environments

- Research walkthrough covering the approach, perception stack, and planning: youtu.be/0HMXJcbxA3w
- Flight demonstration: youtu.be/mjQc9S_NIv8

Robotic Arm Inverse Kinematics Control

- Inverse kinematics with visual feedback, servo driven: youtu.be/yXRN9Aov4zM
- Inverse kinematics, stepper driven: youtube.com/shorts/-wWf4OxTI5E

Line Follower Robot "Thunder" (Champion, DRMC Tech Carnival 2020)

- Short demonstration of the C++ controller software: youtube.com/shorts/ZHi7nnWHWrQ

Kibo Robot Programming Challenge (JAXA and NASA)

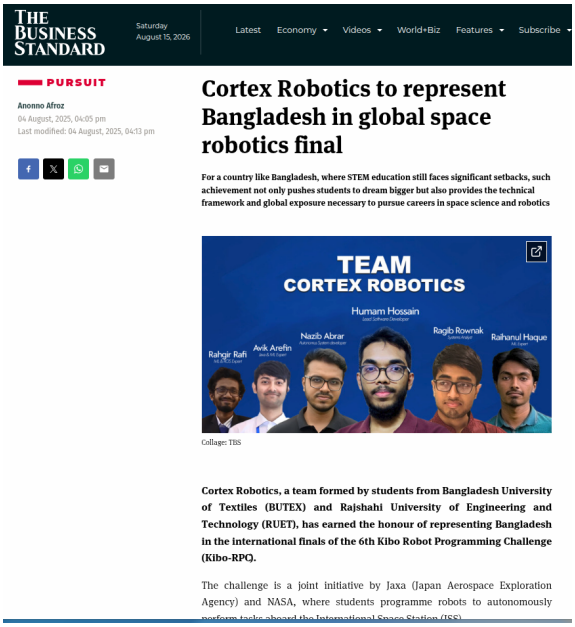
- The Kibo-RPC is hosted by the **Japan Aerospace Exploration Agency (JAXA)** in cooperation with **NASA**. Participating teams develop software that autonomously controls **Astrobee**, NASA's free-flying robot, inside the "Kibo" Japanese Experiment Module aboard the **International Space Station**.
- Competing as **Team Cortex Robotics** (Bangladesh), the team secured first position in the national round and **fourth position globally** in the on-orbit final, completing the mission in 3:42 with three of four correct image matches and correct identification of the target item.
- **On-orbit execution of the team's program aboard the ISS:** youtu.be/sHPJ4LExIgM

JAXA Human Spaceflight Technology Directorate Humans in Space

For Corporate & Research Institute Experiment at Kibo

Rank	Country/Region	Team Name	Mission Completion Time	Number of Correct Matches	Found th correct ite
1	Taiwan	iTron	3:34	4/4	OK
2	Malaysia	Automen	2:45	4/4	NG
3	Philippines	Inflection Point	4:21	4/4	NG
4	Bangladesh	Cortex Robotics	3:42	3/4	OK
5	Indonesia	Narantaka GMAT	3:11	2/4	OK
6	Japan	FUNABASHI	2:42	3/4	NG
7	USA	StarStrikers	Time Over	1/4	NG

Official JAXA final round standings.
Rank 4, Bangladesh, Cortex Robotics.



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PURSUIT

Anonno Afroz
04 August, 2025, 04:05 pm
Last modified: 04 August, 2025, 04:13 pm

Cortex Robotics to represent Bangladesh in global space robotics final

For a country like Bangladesh, where STEM education still faces significant setbacks, such achievement not only pushes students to dream bigger but also provides the technical framework and global exposure necessary to pursue careers in space science and robotics

Collage: TBS

Cortex Robotics, a team formed by students from Bangladesh University of Textiles (BUTEX) and Rajshahi University of Engineering and Technology (RUET), has earned the honour of representing Bangladesh in the international finals of the 6th Kibo Robot Programming Challenge (Kibo-RPC).

The challenge is a joint initiative by Jaxa (Japan Aerospace Exploration Agency) and NASA, where students programme robots to autonomously

The Business Standard, 4 August 2025.
National coverage of the team's qualification.

Verification and Certificates

- **International Rover Design Challenge 2023**. Eleventh position globally with Team Ogrodoot (RUET), issued by the Space Robotics Society.
- **Machine Learning Specialization**. Stanford Online and DeepLearning.AI, September 2023. Verification: coursera.org/verify/specialization/T6GAYS73CU62



International Rover Design Challenge 2023, Space Robotics Society.



Machine Learning Specialization, Stanford Online and DeepLearning.AI.